



Control Charts



Fraction Nonconforming Control Chart:

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

$$\text{Center line} = \bar{p}$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

This Chart is also called P chart



Data for Trial Control Limits, Example 7.1, Sample Size $n = 50$

Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$	Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$
1	12	0.24	17	10	0.20
2	15	0.30	18	5	0.10
3	8	0.16	19	13	0.26
4	10	0.20	20	11	0.22
5	4	0.08	21	20	0.40
6	7	0.14	22	18	0.36
7	16	0.32	23	24	0.48
8	9	0.18	24	15	0.30
9	14	0.28	25	9	0.18
10	10	0.20	26	12	0.24
11	5	0.10	27	7	0.14
12	6	0.12	28	13	0.26
13	17	0.34	29	9	0.18
14	12	0.24	30	6	0.12
15	22	0.44		<u>347</u>	<u>$\bar{p} = 0.2313$</u>
16	8	0.16			

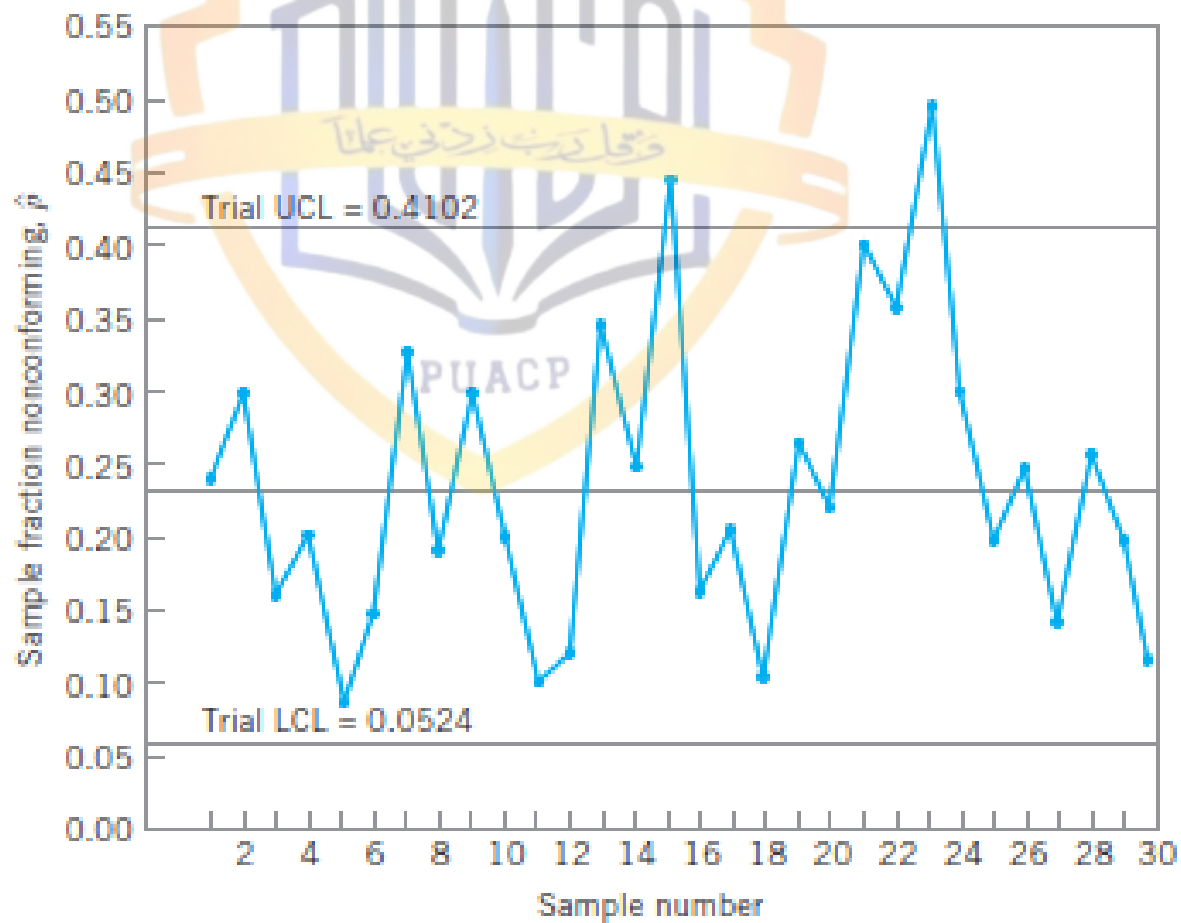
$$\bar{p} = \frac{\sum_{i=1}^m D_i}{mn} = \frac{347}{(30)(50)} = 0.2313$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2313 + 0.1789 = 0.4102$$

and

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2313 - 0.1789 = 0.0524$$





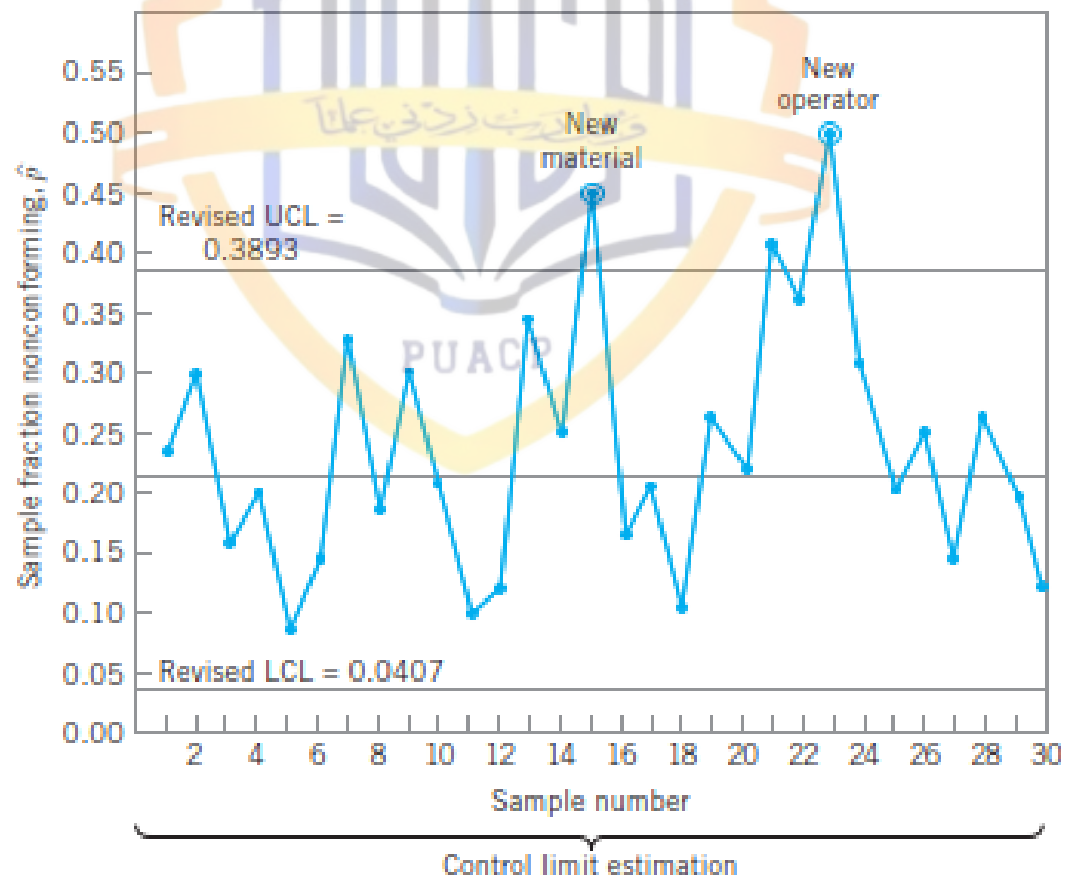
After Eliminating the sample 15 and 23. The new central line and revision control limits are calculated as

$$\bar{p} = \frac{301}{(28)(50)} = 0.2150$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2150 + 3\sqrt{\frac{0.2150(0.7850)}{50}} = 0.3893$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2150 - 3\sqrt{\frac{0.2150(0.7850)}{50}} = 0.0407$$

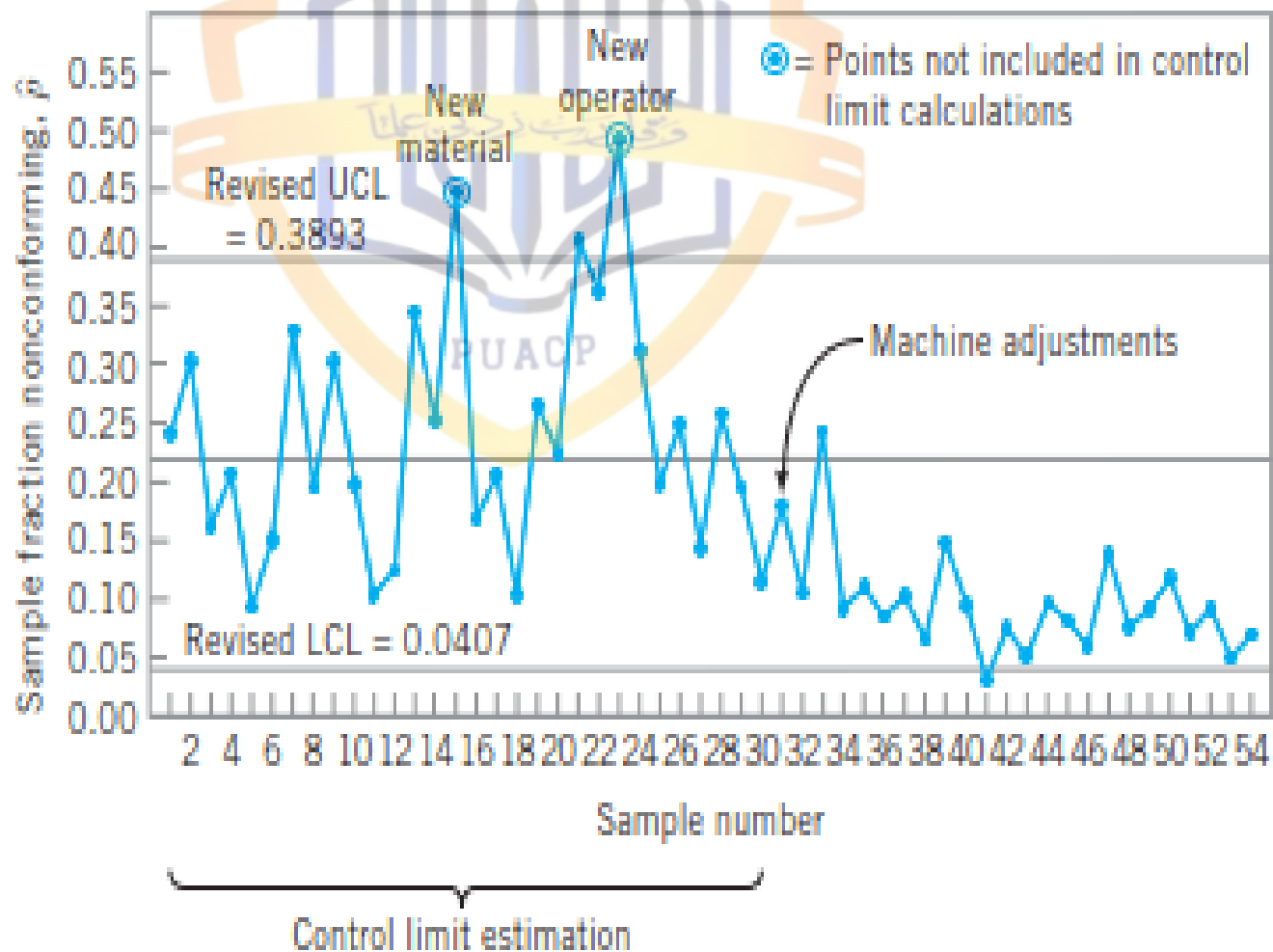




Orange Juice Concentrate Can Data in Samples of Size $n = 50$

Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$	Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$
31	9	0.18	44	6	0.12
32	6	0.12	45	5	0.10
33	12	0.24	46	4	0.08
34	5	0.10	47	8	0.16
35	6	0.12	48	5	0.10
36	4	0.08	49	6	0.12
37	6	0.12	50	7	0.14
38	3	0.06	51	5	0.10
39	7	0.14	52	6	0.12
40	6	0.12	53	3	0.06
41	2	0.04	54	5	0.10
42	4	0.08		133	$\bar{p} = 0.1108$
43	3	0.06			

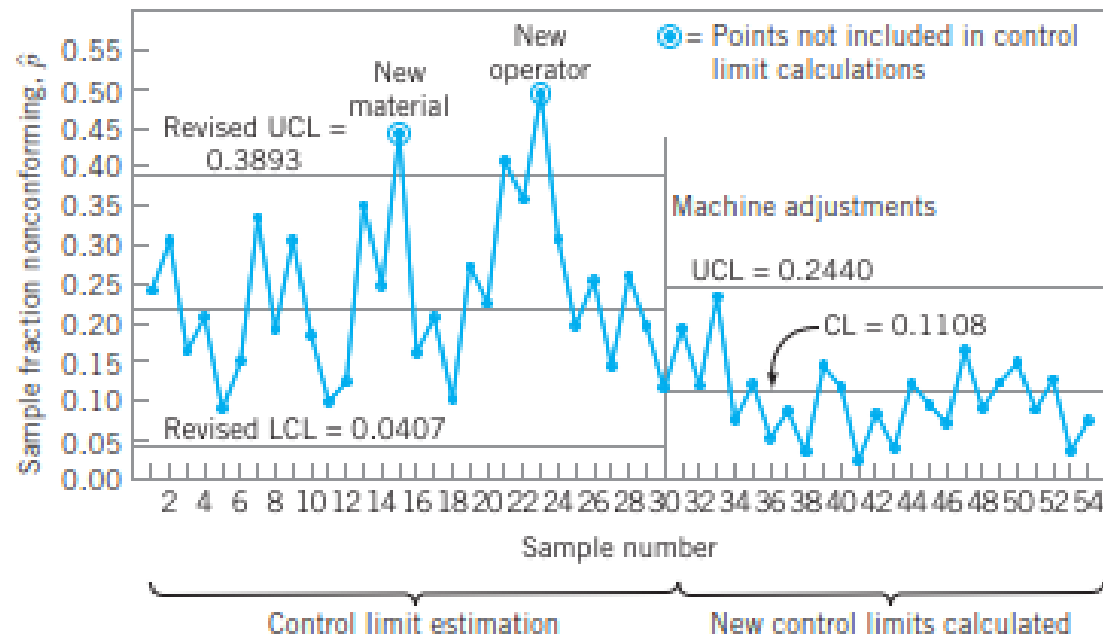




Center line = $\bar{p} = 0.1108$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.1108 + 3\sqrt{\frac{(0.1108)(0.8892)}{50}} = 0.2440$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.1108 - 3\sqrt{\frac{(0.1108)(0.8892)}{50}} = -0.0224 = 0$$



The np Control Chart

$$UCL = np + 3\sqrt{np(1-p)}$$

$$\text{Center line} = np$$

$$LCL = np - 3\sqrt{np(1-p)}$$

Using the data in Table 7.1, we found that

$$p = 0.2313 \quad n = 50$$



Therefore, the parameters of the np control chart would be

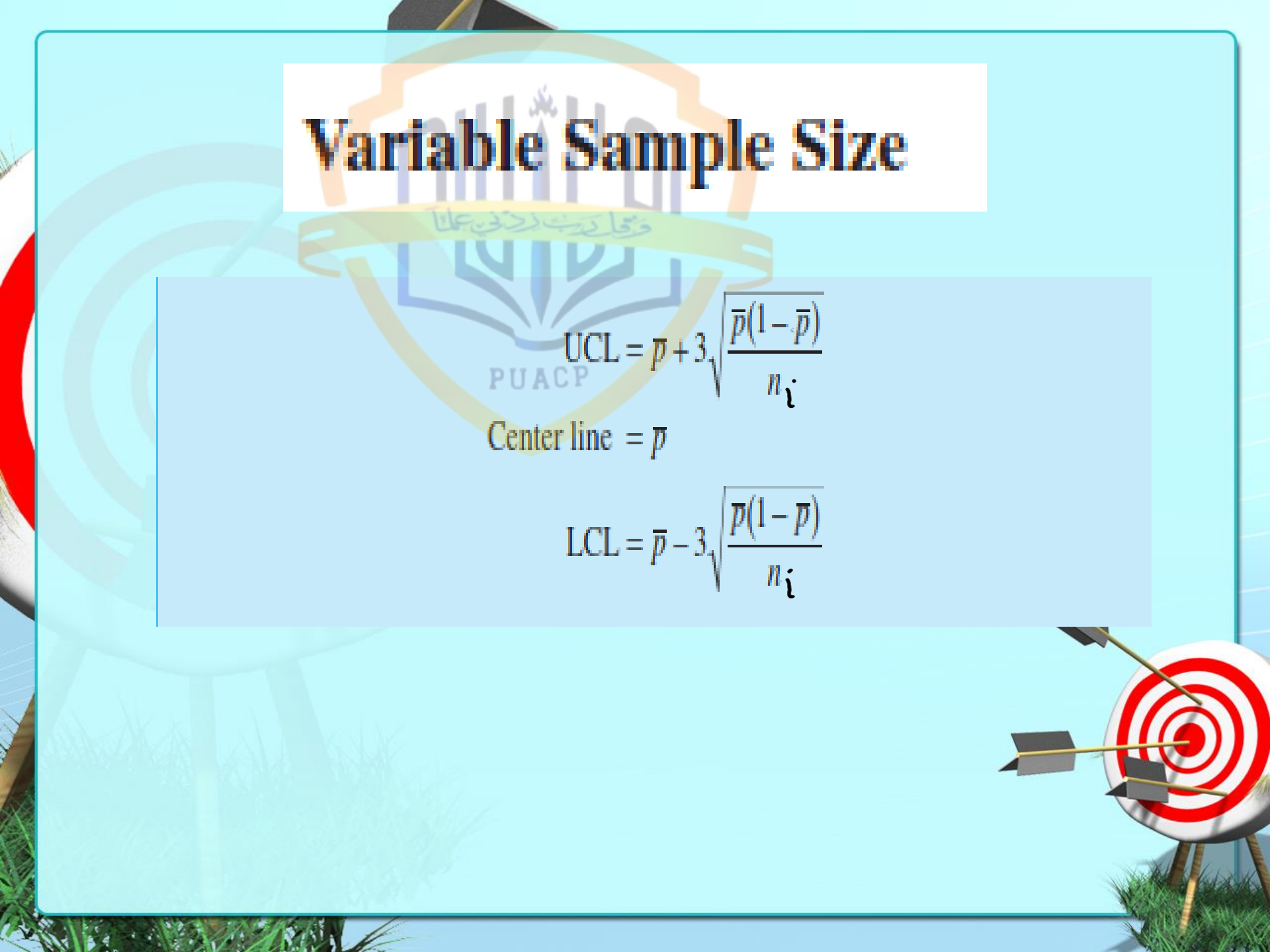
$$\begin{aligned} \text{UCL} &= np + 3\sqrt{np(1-p)} \\ &= 50(0.2313) + 3\sqrt{(50)(0.2313)(0.7687)} \\ &= 20.510 \end{aligned}$$

$$\text{Center line} = np = 50(0.2313) = 11.565$$

$$\begin{aligned} \text{LCL} &= np - 3\sqrt{np(1-p)} \\ &= 50(0.2313) - 3\sqrt{(50)(0.2313)(0.7687)} \\ &= 2.620 \end{aligned}$$



Variable Sample Size


$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$$

$$\text{Center line} = \bar{p}$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$$

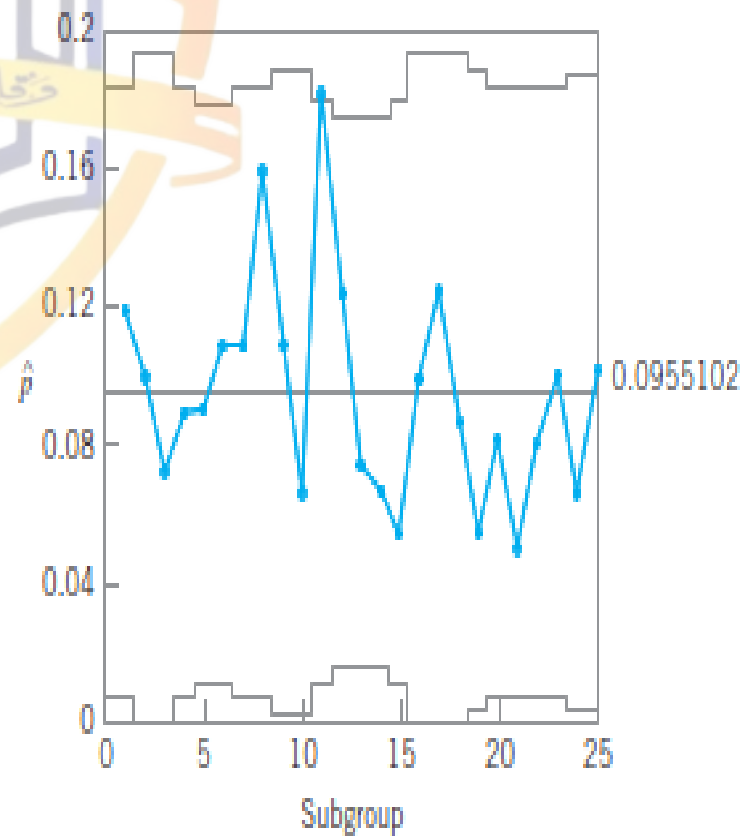
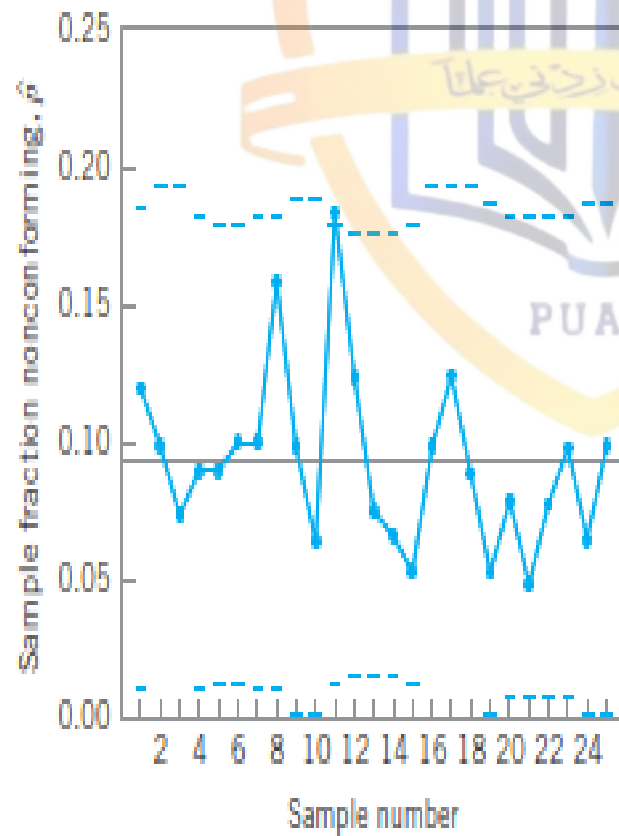
Purchase Order Data for a Control Chart for Fraction Nonconforming with Variable Sample Size

Sample Number, i	Sample Size, n_i	Number of Nonconforming Units, D_i	Sample Fraction Nonconforming, $\hat{p}_i = D_i/n_i$	Standard Deviation	Control Limits	
				$\hat{\sigma}_{\hat{p}} = \sqrt{\frac{(0.096)(0.904)}{n_i}}$	LCL	UCL
1	100	12	0.120	0.029	0.009	0.183
2	80	8	0.100	0.033	0	0.195
3	80	6	0.075	0.033	0	0.195
4	100	9	0.090	0.029	0.009	0.183
5	110	10	0.091	0.028	0.012	0.180
6	110	12	0.109	0.028	0.012	0.180
7	100	11	0.110	0.029	0.009	0.183
8	100	16	0.160	0.029	0.009	0.183
9	90	10	0.110	0.031	0.003	0.189
10	90	6	0.067	0.031	0.003	0.189
11	110	20	0.182	0.028	0.012	0.180
12	120	15	0.125	0.027	0.015	0.177
13	120	9	0.075	0.027	0.015	0.177
14	120	8	0.067	0.027	0.015	0.177
15	110	6	0.055	0.028	0.012	0.180
16	80	8	0.100	0.033	0	0.195
17	80	10	0.125	0.033	0	0.195
18	80	7	0.088	0.033	0	0.195
19	90	5	0.056	0.031	0.003	0.189
20	100	8	0.080	0.029	0.009	0.183
21	100	5	0.050	0.029	0.009	0.183
22	100	8	0.080	0.029	0.009	0.183
23	100	10	0.100	0.029	0.009	0.183
24	90	6	0.067	0.031	0.003	0.189
25	90	9	0.100	0.031	0.003	0.189
	2,450	234	2.383			

$$\bar{p} = \frac{\sum_{i=1}^{25} D_i}{\sum_{i=1}^{25} n_i} = \frac{234}{2,450} = 0.096$$

$$UCL = \bar{p} + 3\hat{\sigma}_{\hat{p}} = 0.096 + 3\sqrt{\frac{(0.096)(0.904)}{n_i}}$$

$$LCL = \bar{p} - 3\hat{\sigma}_{\hat{p}} = 0.096 - 3\sqrt{\frac{(0.096)(0.904)}{n_i}}$$



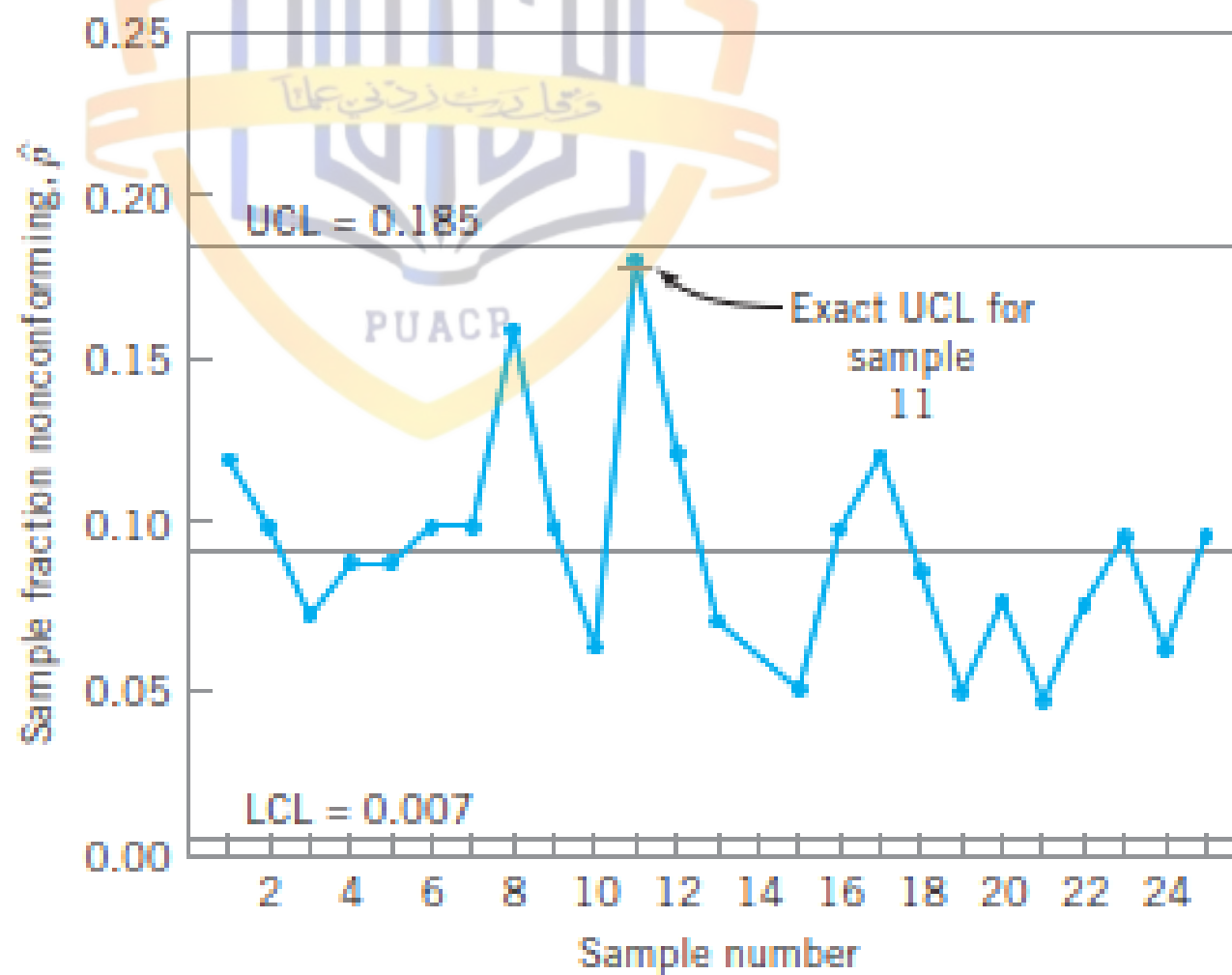
Control Limits Based on an Average Sample Size.

$$\bar{n} = \frac{\sum_{i=1}^{25} n_i}{25} = \frac{2,450}{25} = 98$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} = 0.096 + 3\sqrt{\frac{(0.096)(0.904)}{98}} = 0.185$$

and

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} = 0.096 - 3\sqrt{\frac{(0.096)(0.904)}{98}} = 0.007$$



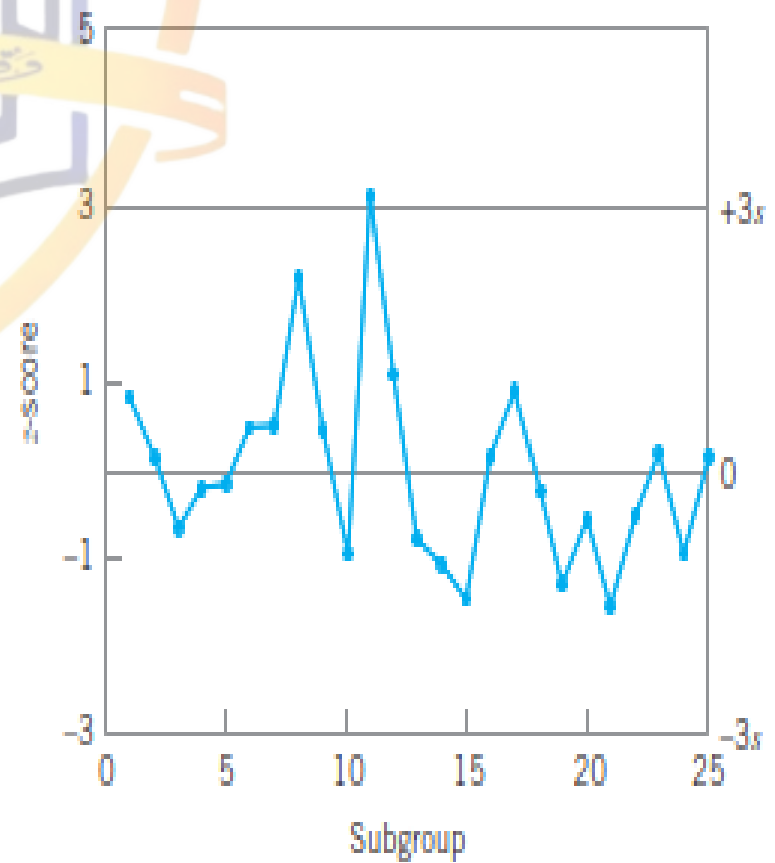
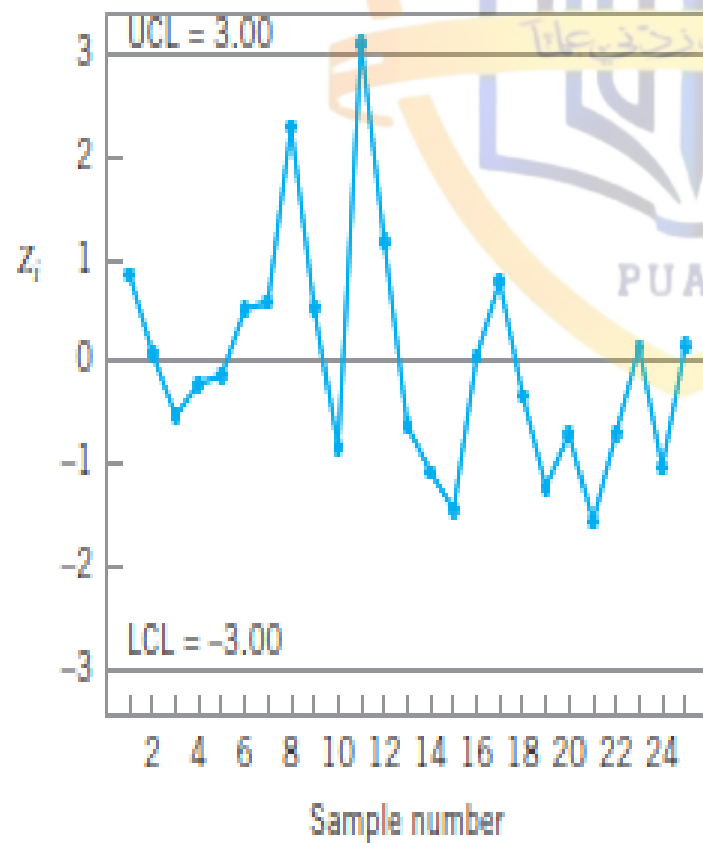
The Standardized Control Chart.

$$Z_i = \frac{\hat{p}_i - p}{\sqrt{\frac{p(1-p)}{n_i}}}$$



Calculations for the Standardized Control Chart in Figure 7.9, $\hat{p} = 0.096$

Sample Number, i	Sample Size, n_i	Number of Nonconforming Units, D_i	Sample Fraction Nonconforming, $\hat{p}_i = D_i/n_i$	Standard Deviation $\hat{\sigma}_p = \sqrt{\frac{(0.096)(0.904)}{n_i}}$	$z_i = \frac{\hat{p}_i - \bar{p}}{\sqrt{\frac{(0.096)(0.904)}{n_i}}}$
1	100	12	0.120	0.029	0.83
2	80	8	0.100	0.033	0.12
3	80	6	0.075	0.033	-0.64
4	100	9	0.090	0.029	-0.21
5	110	10	0.091	0.028	-0.18
6	110	12	0.109	0.028	0.46
7	100	11	0.110	0.029	0.48
8	100	16	0.160	0.029	2.21
9	90	10	0.110	0.031	0.45
10	90	6	0.067	0.031	-0.94
11	110	20	0.182	0.028	3.07
12	120	15	0.125	0.027	1.07
13	120	9	0.075	0.027	-0.78
14	120	8	0.067	0.027	-1.07
15	110	6	0.055	0.028	-1.46
16	80	8	0.100	0.033	0.12
17	80	10	0.125	0.033	0.88
18	80	7	0.088	0.033	-0.24
19	90	5	0.056	0.031	-1.29
20	100	8	0.080	0.029	-0.55
21	100	5	0.050	0.029	-1.59
22	100	8	0.080	0.029	-0.55
23	100	10	0.100	0.029	0.14
24	90	6	0.067	0.031	-0.94
25	90	9	0.100	0.031	0.13



The Operating-Characteristic Function and Average Run Length Calculations

$$\beta = P\{\hat{p} < \text{UCL} | p\} - P\{\hat{p} \leq \text{LCL} | p\}$$

$$= P\{D < n\text{UCL} | p\} - P\{D \leq n\text{LCL} | p\}$$

$$\beta = P\{D < (50)(0.3697) | p\} - P\{D \leq (50)(0.0303) | p\}$$

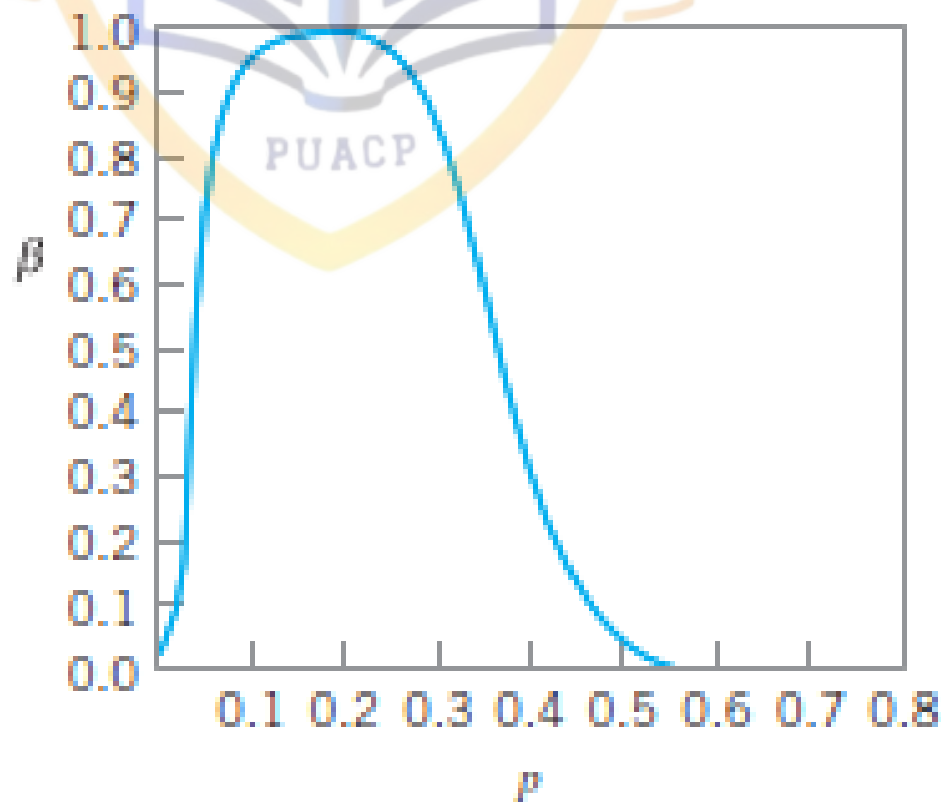
$$= P\{D < 18.49 | p\} - P\{D \leq 1.52 | p\}$$

$$\beta = P\{D \leq 18 | p\} - P\{D \leq 1 | p\}$$

Calculations^a for Constructing the OC Curve for a Control Chart for Fraction Nonconforming with $n = 50$, $\text{LCL} = 0.0303$, and $\text{UCL} = 0.3697$

p	$P\{D \leq 18 p\}$	$P\{D \leq 1 p\}$	$\beta = P\{D \leq 18 p\} - P\{D \leq 1 p\}$
0.01	1.0000	0.9106	0.0894
0.03	1.0000	0.5553	0.4447
0.05	1.0000	0.2794	0.7206
0.10	1.0000	0.0338	0.9662
0.15	0.9999	0.0029	0.9970
0.20	0.9975	0.0002	0.9973
0.25	0.9713	0.0000	0.9713
0.30	0.8594	0.0000	0.8594
0.35	0.6216	0.0000	0.6216
0.40	0.3356	0.0000	0.3356
0.45	0.1273	0.0000	0.1273
0.50	0.0325	0.0000	0.0325
0.55	0.0053	0.0000	0.0053

"The probabilities in this table were found by evaluating the cumulative binomial distribution. For small p ($p < 0.1$, say) the Poisson approximation could be used, and for larger values of p the normal approximation could be used.



Thus, if the process is in control, ARL_0 is

$$ARL_0 = \frac{1}{\alpha}$$

and if it is out of control, then

$$ARL_1 = \frac{1}{1 - \beta}$$



These probabilities (α , β) can be calculated directly from the binomial distribution or read from an OC curve.

To illustrate, consider the control chart for fraction nonconforming used in the OC curve calculations in Table 7.6. This chart has parameters $n = 50$, $UCL = 0.3697$, $LCL = 0.0303$, and the center line is $\bar{p} = 0.20$. From Table 7.6 (or the OC curve in Fig. 7.11) we find that if the process is in control with $p = \bar{p}$, the probability of a point plotting in control is 0.9973. Thus, in this case $\alpha = 1 - \beta = 0.0027$, and the value of ARL_0 is

$$1 - \alpha = 0.9973$$

$$\alpha = 0.0027$$

$$ARL_0 = \frac{1}{\alpha} = \frac{1}{0.0027} \approx 370$$

Therefore, if the process is really in control, we will experience a false out-of-control signal about every 370 samples. (This will be approximately true, in general, for any Shewhart control chart with three-sigma limits.) This in-control ARL_0 is generally considered to be satisfactorily large. Now suppose that the process shifts out of control to $p = 0.3$. Table 7.6 indicates that if $p = 0.3$, then $\beta = 0.8594$. Therefore, the value of ARL_1 is

$$ARL_1 = \frac{1}{1 - \beta} = \frac{1}{1 - 0.8594} \approx 7$$

Control Charts for Nonconformities (Defects)

Control Chart for Nonconformities:

PUACP

$$UCL = \bar{c} + 3\sqrt{\bar{c}}$$

$$\text{Center line} = \bar{c}$$

$$LCL = \bar{c} - 3\sqrt{\bar{c}}$$



Data on the Number of Nonconformities in Samples of 100 Printed Circuit Boards

Sample Number	Number of Nonconformities	Sample Number	Number of Nonconformities
1	21	14	19
2	24	15	10
3	16	16	17
4	12	17	13
5	15	18	22
6	5	19	18
7	28	20	39
8	20	21	30
9	31	22	24
10	25	23	16
11	20	24	19
12	24	25	17
13	16	26	15

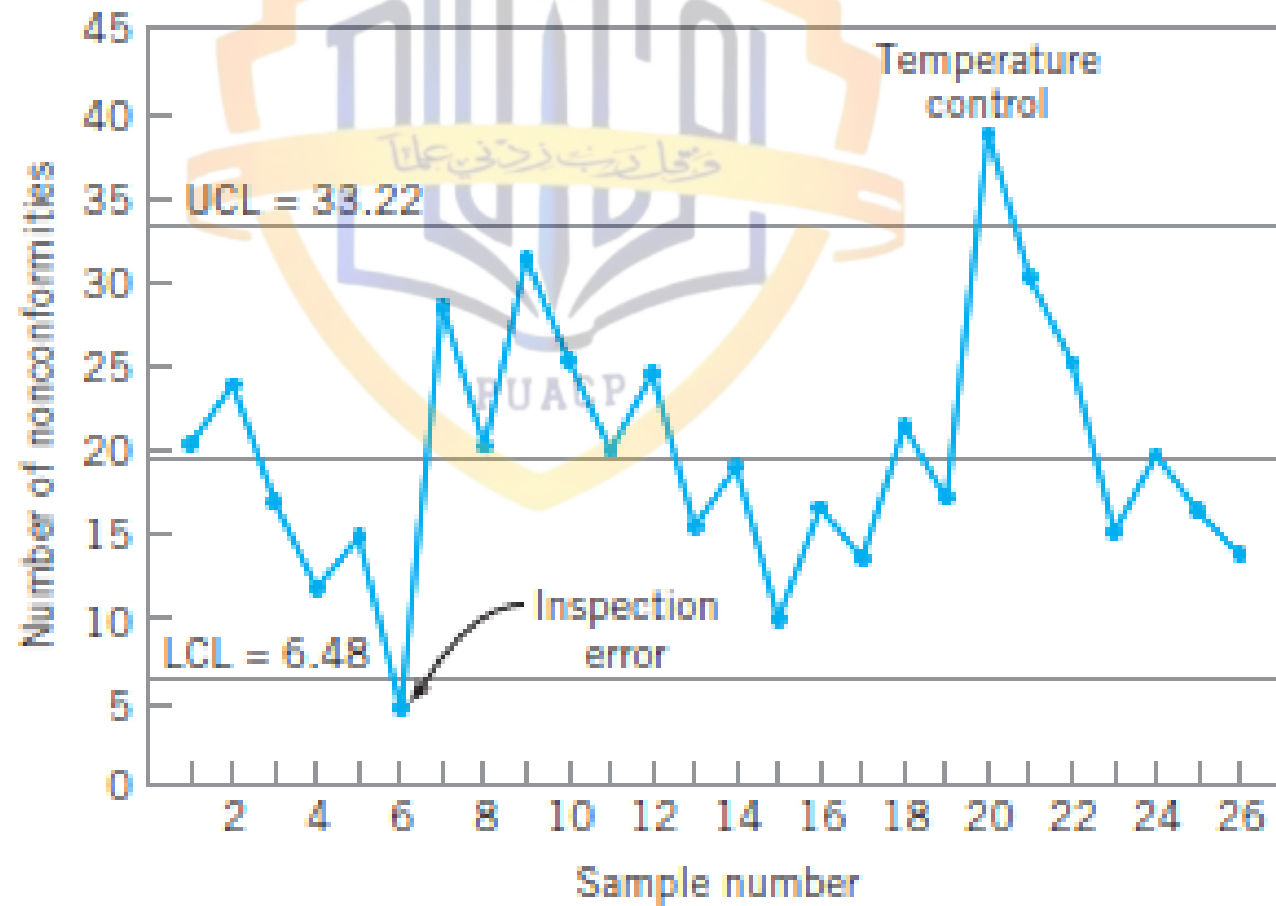
$$\bar{c} = \frac{516}{26} = 19.85$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}} = 19.85 + 3\sqrt{19.85} = 33.22$$

$$\text{Center line} = \bar{c} = 19.85$$

$$LCL = \bar{c} - 3\sqrt{\bar{c}} = 19.85 - 3\sqrt{19.85} = 6.48$$





$$\bar{c} = \frac{472}{24} = 19.67$$

and the revised control limits are

$$UCL = \bar{c} + 3\sqrt{\bar{c}} = 19.67 + 3\sqrt{19.67} = 32.97$$

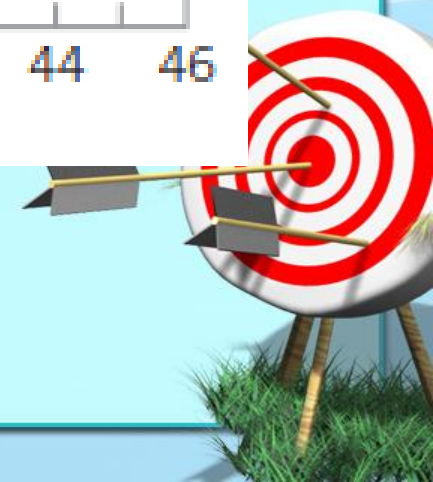
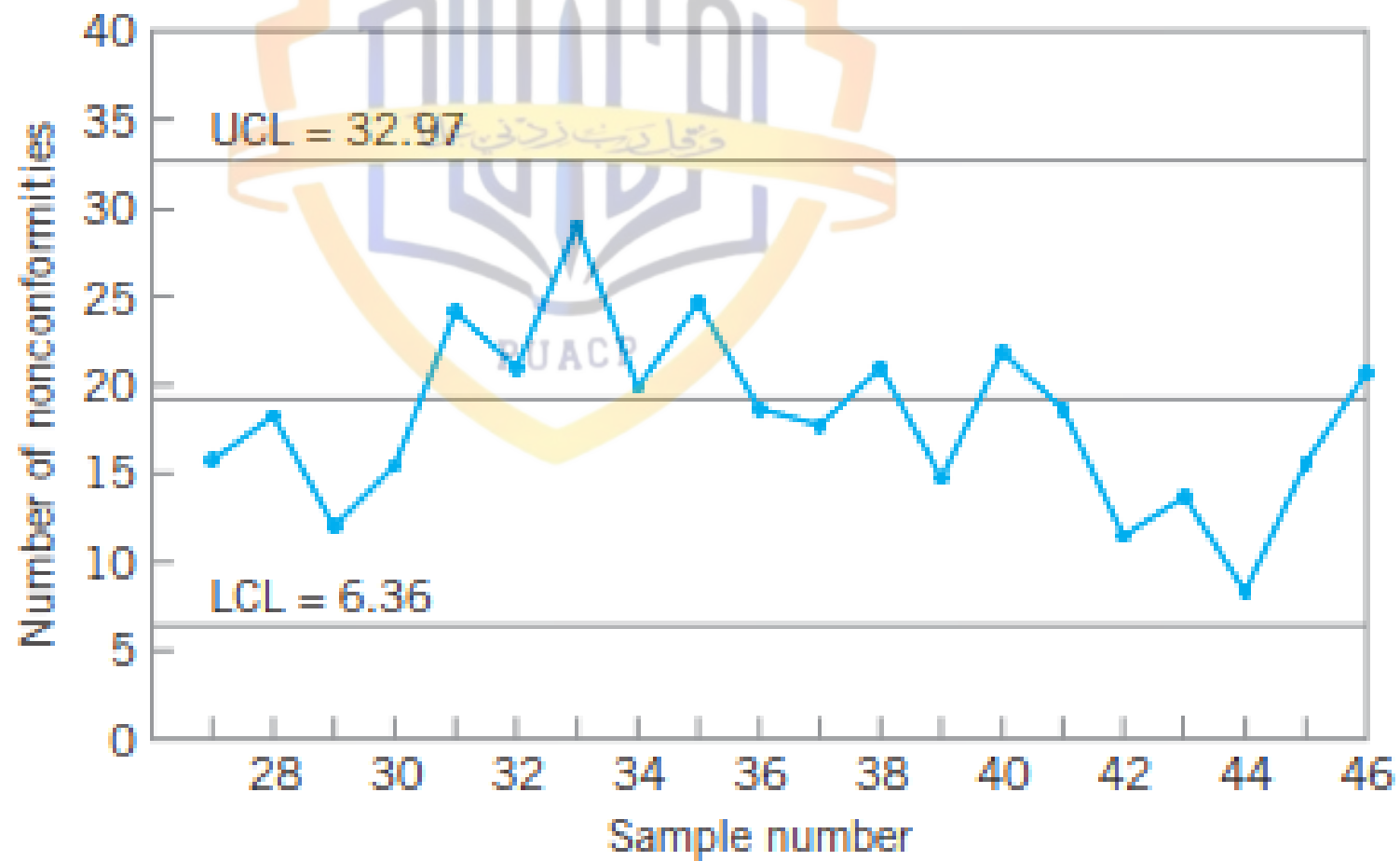
$$\text{Center line} = \bar{c} = 19.67$$

$$LCL = \bar{c} - 3\sqrt{\bar{c}} = 19.67 - 3\sqrt{19.67} = 6.36$$

Additional Data for the Control Chart for Nonconformities, Example 7.3

Sample Number	Number of Nonconformities	Sample Number	Number of Nonconformities
27	16	37	18
28	18	38	21
29	12	39	16
30	15	40	22
31	24	41	19
32	21	42	12
33	28	43	14
34	20	44	9
35	25	45	16
36	19	46	21





The $\bar{u}$ Chart.

Control Chart for Average Number of
Nonconformities per Unit

PUACP

$$UCL = \bar{u} + 3\sqrt{\frac{\bar{u}}{n}}$$

$$\text{Center line} = \bar{u}$$

$$LCL = \bar{u} - 3\sqrt{\frac{\bar{u}}{n}}$$



Data on Number of Shipping Errors in a Supply Chain Network

Sample Number (week), i	Sample Size, n	Total Number of Errors (Nonconformities), x_i	Average Number of Errors (Nonconformities) per Unit, $u_i = x_i/n$
1	50	2	0.04
2	50	3	0.06
3	50	8	0.16
4	50	1	0.02
5	50	1	0.02
6	50	4	0.08
7	50	1	0.02
8	50	4	0.08
9	50	5	0.10
10	50	1	0.02
11	50	8	0.16
12	50	2	0.04
13	50	4	0.08
14	50	3	0.06
15	50	4	0.08
16	50	1	0.02
17	50	8	0.16
18	50	3	0.06
19	50	7	0.14
20	50	4	0.08
		74	1.48



$$\bar{p} = \frac{\sum_{i=1}^{20} u_i}{20} = \frac{1.48}{20} = 0.0740$$

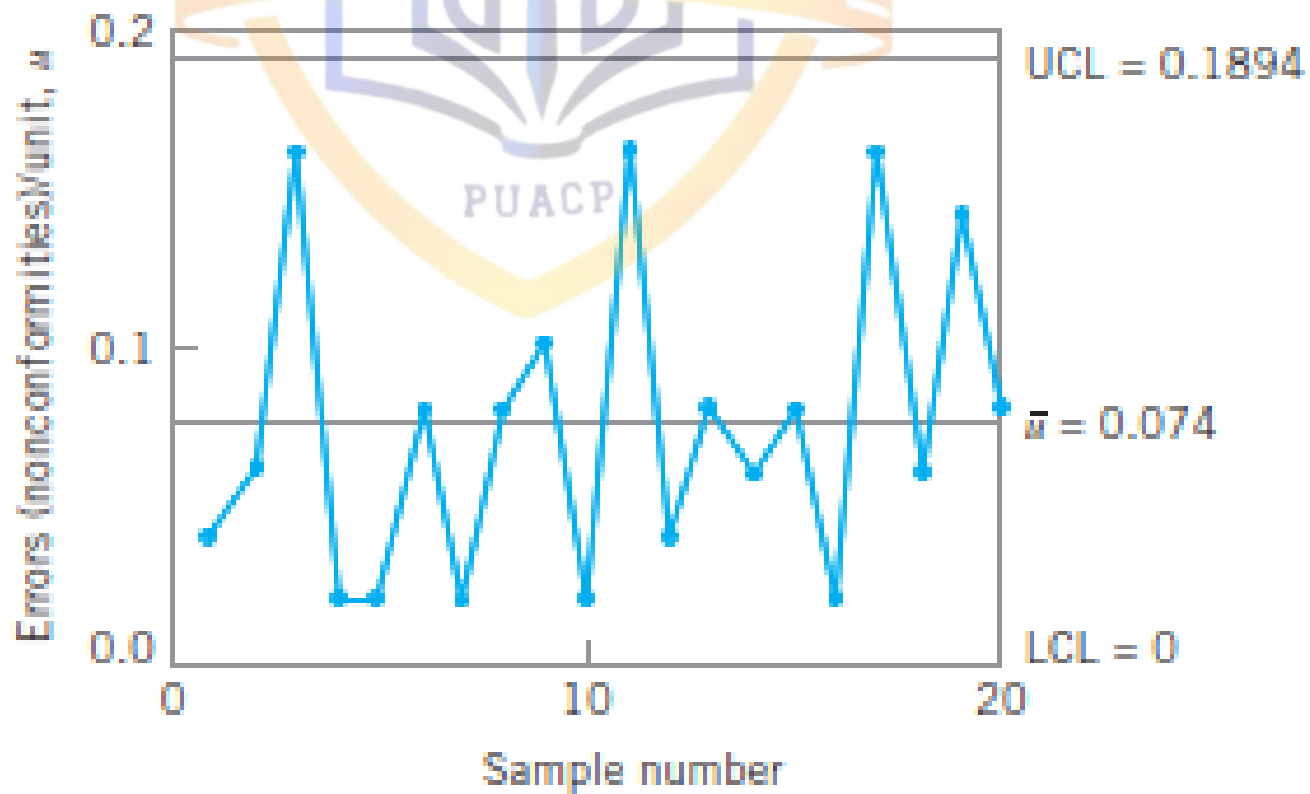
Therefore, the parameters of the control chart are

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}}{n}} = 0.0740 + 3\sqrt{\frac{0.0740}{50}} = 0.1894$$

$$\text{Center line} = \bar{p} = 1.93$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}}{n}} = 0.0740 - 3\sqrt{\frac{0.0740}{50}} = -0.0414$$





The u Chart.

For Variable Sample Size

$$UCL = \bar{u} + 3\sqrt{\frac{\bar{u}}{n_i}}$$

$$\text{Center line} = \bar{u}$$

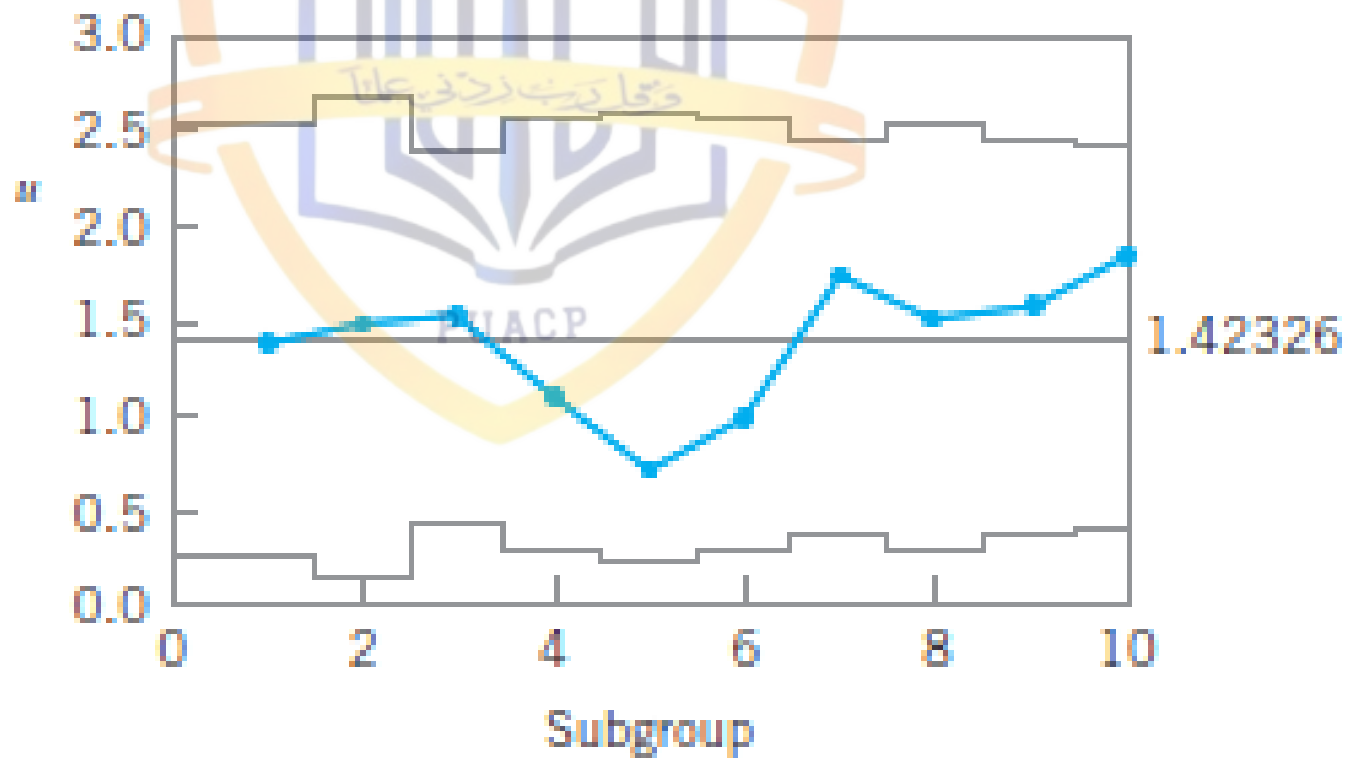
$$LCL = \bar{u} - 3\sqrt{\frac{\bar{u}}{n_i}}$$



$$\bar{u} = \frac{153}{107.5} = 1.42$$

Calculation of Control Limits, Example 7.5

Roll Number, i	n_i	$UCL = \bar{u} + 3\sqrt{\bar{u}/n_i}$	$LCL = \bar{u} - 3\sqrt{\bar{u}/n_i}$
1	10.0	2.55	0.29
2	8.0	2.68	0.16
3	13.0	2.41	0.43
4	10.0	2.55	0.29
5	9.5	2.58	0.26
6	10.0	2.55	0.29
7	12.0	2.45	0.39
8	10.5	2.52	0.32
9	12.0	2.45	0.39
10	12.5	2.43	0.41



The Standardized Control Chart.

Use a standardized control chart (this is the preferred option). This second alternative would involve plotting a standardized statistic

$$Z_i = \frac{u_i - \bar{u}}{\sqrt{\frac{\bar{u}}{n_i}}}$$





**THANK
YOU**